PHYSIOLOGY · MCQ PRACTICE

GUE BLOCK

Physiology of Urine Formation — MCQ Practice

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Physiology GUE Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Covers glomerular filtration, GFR, Starling pressures, factors affecting GFR, tubular reabsorption and secretion.

Q Questions

1. What is the correct order of the three processes of urine formation? EASY

A. Reabsorption → Filtration → Secretion

B. Filtration → Reabsorption → Secretion

C. Secretion → Filtration → Reabsorption

D. Filtration → Secretion → Reabsorption

E. Reabsorption → Secretion → Filtration

2. Approximately what fraction of cardiac output enters the kidneys as renal blood flow? EASY

A. 5%

B. 10%

C. 26%

D. 50%

E. 75%

3. What is the normal glomerular filtration rate (GFR)? EASY

A. ~12.5 mL/min

B. ~60 mL/min

C. ~125 mL/min

D. ~500 mL/min

E. ~1300 mL/min

4. Glomerular filtration is called "ultrafiltration" because: EASY

A. It only filters red blood cells

B. Even minute particles are filtered, but plasma proteins are excluded by size

C. It filters nothing larger than water

D. Filtration only happens once per day

E. It uses active transport to filter particles

5. A patient's glomerular capillary pressure is 60 mmHg, colloidal osmotic pressure is 25 mmHg, and Bowman's capsule hydrostatic pressure is 15 mmHg. What is the net filtration pressure? MEDIUM

A. 100 mmHg

B. 60 mmHg

C. 40 mmHg

D. 20 mmHg

E. 10 mmHg

6. If GFR is 125 mL/min and net filtration pressure is 20 mmHg, what is the filtration coefficient? MEDIUM

A. 0.16 mL/min/mmHg
B. 2.5 mL/min/mmHg
C. 6.25 mL/min/mmHg
D. 25 mL/min/mmHg
E. 145 mL/min/mmHg

7. Which of the three Starling forces determining GFR is the only one that FAVORS filtration? MEDIUM

A. Colloidal osmotic pressure in the glomeruli
B. Hydrostatic pressure in Bowman's capsule
C. Glomerular capillary pressure
D. Tubular hydrostatic pressure in the collecting duct
E. Oncotic pressure in the renal vein

8. A patient has a tumor compressing the ureter, raising pressure throughout the urinary tract including Bowman's capsule. What happens to GFR? MEDIUM

A. GFR rises because capsular pressure favors filtration
B. GFR falls because capsular hydrostatic pressure opposes filtration and NFP drops
C. GFR is unaffected since capsular pressure does not enter the NFP equation
D. GFR rises because filtration fraction increases automatically
E. Glomerular capillary pressure falls to compensate, leaving GFR unchanged

9. Which substances are essentially completely reabsorbed from the renal tubule under normal conditions? MEDIUM

A. Urea and uric acid
B. Metabolic waste products
C. Glucose and amino acids
D. Creatinine and drugs
E. Excess ions only

10. By what mechanism are glucose and amino acids reabsorbed in the proximal convoluted tubule? MEDIUM

A. Simple diffusion down a concentration gradient
B. Secondary active transport via sodium cotransport
C. Bulk flow through tight junctions only
D. Primary active transport using the H^+-ATPase
E. Facilitated diffusion independent of sodium

11. Which route of reabsorption involves substances moving through the tight junctions between tubular cells rather than through the cells themselves? MEDIUM

A. Transcellular route
B. Paracellular route
C. Active transport route
D. Osmotic route
E. Secretory route

12. In which nephron segment is bicarbonate reabsorbed, and in what form? MEDIUM

A. Loop of Henle, as free $HCO3^-$ ions
B. Mostly the proximal tubule, in the form of CO2
C. Only the collecting duct, as carbonic acid
D. Distal tubule, by simple diffusion only
E. Nowhere — bicarbonate is always excreted

13. Which of the following is secreted into the proximal convoluted tubule? MEDIUM

A. Potassium only
B. Ammonia
C. Urea
D. Calcium
E. Bicarbonate

14. Which nephron segment secretes urea into the tubular fluid? MEDIUM

A. Proximal convoluted tubule
B. Loop of Henle
C. Distal convoluted tubule only
D. Collecting duct only
E. Bowman's capsule

15. A substance is actively reabsorbed from the tubule, and at a certain filtered load it begins appearing in the urine even though active transporters are still present and functioning. This finding is best explained by: HARD

A. Filtration fraction has reached zero
B. The substance has exceeded its tubular transport maximum (Tm)
C. Glomerular capillary pressure has fallen to zero
D. The paracellular route has been blocked
E. Secretion has converted to reabsorption

16. A 32-year-old has a selective genetic defect preventing synthesis of the Na^+-glucose cotransporter (SGLT) in the proximal tubule, with normal GFR and normal blood glucose. Urinalysis would most likely show: HARD

A. No glucose in the urine, since GFR is normal
B. Glucosuria despite a normal blood glucose level, because filtered glucose cannot be reabsorbed
C. Glucosuria only during exercise
D. Ketonuria without glucosuria
E. Proteinuria from compensatory glomerular leak

17. Two patients both have GFR of 125 mL/min. Patient A has glomerular capillary pressure 60 mmHg, colloidal osmotic pressure 25 mmHg, capsular pressure 15 mmHg. Patient B has the same glomerular and capsular pressures, but a disease state has doubled their colloidal osmotic pressure to 50 mmHg. What happens to Patient B's NFP compared to Patient A's? HARD

A. Unchanged, since colloidal osmotic pressure does not affect NFP
B. Falls from 20 mmHg to −5 mmHg, meaning filtration is opposed overall
C. Rises to 40 mmHg
D. Falls to 0 mmHg exactly and stays there indefinitely
E. Rises because colloidal osmotic pressure favors filtration

18. A patient has isolated loss of peritubular capillary function in the PCT, impairing bulk flow out of the interstitium into the capillary, while the apical (luminal) membrane transporters and tight junctions remain intact. Which step of substance movement is most directly disrupted? HARD

A. Filtration at the glomerulus
B. Movement from tubular lumen into the tubular cell
C. Movement from interstitial fluid into the peritubular capillary — the final step shared by both transcellular and paracellular routes
D. Secretion of potassium by the Na-K pump
E. Tubular transport maximum for glucose

19. A patient has selective destruction of the macula densa, impairing tubuloglomerular feedback, while all other GFR-determining factors remain normal. During a high-NaCl tubular load, what is the most likely immediate consequence? HARD

A. GFR is permanently fixed at the current value regardless of tubular NaCl, since macula densa feedback is the sole regulator of renal blood flow

B. The kidney loses one of its mechanisms for matching glomerular blood flow and GFR to tubular NaCl delivery, though other factors (e.g. systemic pressure, sympathetic tone) still influence GFR

C. Glomerular capillary pressure immediately rises to 120 mmHg

D. Bowman's capsule hydrostatic pressure falls to zero

E. Filtration fraction becomes undefined

20. Compare two substances: Substance X is reabsorbed by secondary active transport coupled to Na^+ in the PCT (like glucose); Substance Y is secreted into the PCT (like ammonia). If a drug selectively poisons the basolateral Na^+/K^+-ATPase in proximal tubular cells (destroying the Na^+ gradient that powers secondary active transport), which is more directly affected? HARD

A. Substance Y's secretion, since secretion always requires the Na+/K+-ATPase directly

B. Substance X's reabsorption, since it depends on the Na^+ gradient the Na^+/K^+-ATPase maintains for secondary active transport; Substance Y's secretion is not described as depending on this Na^+-coupled mechanism

C. Neither — both are independent of any sodium gradient

D. Both are equally affected because they occur in the same tubular segment

E. Only glomerular filtration is affected, not tubular transport

✓ Answer key

Q	Answer	Difficulty	Why
1	**B**	EASY	Blood is filtered at the glomerulus first, then reabsorption and secretion modify the filtrate along the tubule.
2	**C**	EASY	Normally ~1300 mL/min of blood enters the kidneys, about 26% of cardiac output.
3	**C**	EASY	Normal GFR is ~125 mL/min, equal to about 180 L of filtrate per day from both kidneys.
4	**B**	EASY	Even minute particles pass, but plasma proteins are too large for the capillary filtration pores and remain in the blood.
5	**D**	MEDIUM	NFP = 60 − (25+15) = 20 mmHg, the normal value (range 15–20 mmHg).
6	**C**	MEDIUM	Filtration coefficient = GFR ÷ NFP = 125 ÷ 20 = 6.25 mL/min/mmHg.
7	**C**	MEDIUM	Glomerular capillary pressure (~60 mmHg) favors filtration; the other two listed Starling forces oppose it.
8	**B**	MEDIUM	Capsular hydrostatic pressure opposes filtration; raising it lowers NFP and therefore GFR.
9	**C**	MEDIUM	Essential substances like glucose and amino acids are completely reabsorbed in the PCT; unwanted substances like metabolic wastes are not reabsorbed and are excreted.
10	**B**	MEDIUM	Glucose and amino acids are reabsorbed by secondary active transport, cotransported with sodium.
11	**B**	MEDIUM	The paracellular route moves substances through the intercellular space via tight junctions; the transcellular route moves substances through the cell.
12	**B**	MEDIUM	Bicarbonate is reabsorbed actively, mostly in the proximal tubule, in the form of carbon dioxide.
13	**B**	MEDIUM	Ammonia is secreted in the PCT; H+ is secreted in PCT and DCT; potassium is secreted in PCT, DCT and CD; urea is secreted in the loop of Henle.
14	**B**	MEDIUM	The lecture specifies urea is secreted in the loop of Henle, distinct from potassium, ammonia, and H+ secretion sites.
15	**B**	HARD	For every actively reabsorbed substance there is a maximum rate (Tm) at which it can be reabsorbed; exceeding it causes the excess to appear in urine.
16	**B**	HARD	Glucose reabsorption in the PCT depends on the Na^+-glucose cotransporter (secondary active transport); without it, normally-filtered glucose stays in the tubule and appears in urine despite normal plasma glucose.
17	**B**	HARD	

			NFP = capillary pressure − (colloidal osmotic + capsular). Doubling the opposing colloidal osmotic pressure to 50 gives NFP = 60 − (50+15) = −5 mmHg — filtration is no longer favored.
18	**C**	HARD	Both the transcellular and paracellular routes converge on the same final step: movement from interstitial fluid into the peritubular capillary. Disrupting peritubular capillary uptake blocks this shared final step regardless of which route substances took to reach the interstitium.
19	**B**	HARD	Tubuloglomerular feedback via the macula densa is one of several factors affecting GFR; losing it removes that specific feedback loop but other regulators (renal blood flow, systemic pressure, sympathetic stimulation, hormones) continue to act on GFR.
20	**B**	HARD	Secondary active transport (glucose/amino acid reabsorption) depends on the Na^+ gradient generated by the Na^+/K^+-ATPase; poisoning this pump collapses that gradient and directly impairs Na^+-coupled reabsorption, whereas the lecture describes secretion (e.g. ammonia) through a separate mechanism.